

MSDM 5056 Network Modeling
Assignment 3 (due 5th November, 2025)

Submit your assignment solution on Canvas. You may discuss with others or seek help from your TA, but you should not directly copy from others. Otherwise, it will be considered as plagiarism.

(1) Degree distribution

Consider a random network $G(n; p)$ with $n = 4 \times 10^6$ nodes and linking probability $p = 10^{-4}$.

- (a) Calculate the mean degree c of the network.
- (b) Calculate the standard deviation σ of degree using the approximated degree distribution discussed in the lecture notes. (*Refer to p.20 of Lecture 6*)
- (c) Assume that you have observed a node with degree 2×10^3 . How many standard deviations is this observation from the mean? Is this an expected observation or is this an unexpected observation? (*Refer to p.21 of Lecture 6*)

Solution:

- (a) The mean degree is $c = p(n - 1) = 10^{-4} \times (4 \times 10^6 - 1) \approx 400$.
- (b) The degree distribution $P(k)$ can be approximated with a Poisson distribution with mean degree $\langle k \rangle = c = 400$, so the standard deviation is $\sigma = \sqrt{c} = 20$.
- (c) This observation is unexpected because $\frac{|2000 - c|}{\sigma} = \frac{2000 - 400}{20} = 80$.

(2) Clustering coefficient

Consider a random network $G(n; m)$ with $n \rightarrow \infty$ nodes and m edges. It has a constant mean degree $c = 2m/n$. Assume that you do not know the value of its clustering coefficient C .

- (a) Show that the expected number of triangles in the network is $c^3/6$.
- (b) Show that the expected number of connected triples in the network is $nc^2/2$.
- (c) Hence, derive the clustering coefficient C and confirm that it agrees with the expression on p.19 in Lecture 6.

Solution:

- (a) There are a total of $n(n - 1)(n - 2)/3!$ ways to pick any three nodes in the network. The probability that any two of the three nodes are neighbors is given by $c/(n - 1)$. The probability that the three nodes can form a triangle is therefore

given by

$$\frac{n(n-1)(n-2)}{3!} \left(\frac{c}{n-1} \right)^3 \approx \frac{c^3}{6}$$

in the large n limit.

- (b) We again have a total of $n(n-1)(n-2)/3!$ ways to pick any three nodes in the network and a probability of $c/(n-1)$ for any of the two nodes to be neighbors. There are 3 ways to pick any two out of the three nodes. Putting these things altogether, one obtains

$$\frac{n(n-1)(n-2)}{3!} \left(\frac{c}{n-1} \right)^2 \times 3 \approx \frac{nc^2}{2}$$

connected triples in the network.

- (c) Dividing three times the number of triangles obtained in (a) by the number of connected triples in (b) gives

$$3 \times \frac{c^3}{6} \div \frac{nc^2}{2} \approx \frac{c}{n}$$

as expected.

(3) Generating function and excess degree

Consider a random network $G(n; p)$ with $n \rightarrow \infty$ nodes and linking probability p . Letting $c = p(n-1)$, the network approximately has a Poisson degree distribution

$$P_{\text{Poisson}}(k) = \frac{1}{k!} c^k e^{-c}$$

- (a) Calculate the generating function

$$G_0(x) = \sum_{k=0}^{\infty} P_{\text{Poisson}}(k) x^k$$

of a Poisson degree distribution.

- (b) Using the properties of a generating function, evaluate $\langle k \rangle$ and $\langle k(k-1) \rangle$ of a Poisson degree distribution, where $\langle \dots \rangle$ denotes the expected value of the quantity inside the brackets.

In many real-world calculations, we often need to go along an edge of a node s to reach another node t . This makes us less interested in the total degree of t than in its *excess degree*, which is the number of edges attached to t excluding the one we arrived along from s . Therefore, a node's excess degree is just one less than its total degree. When a network has degree distribution $P(k)$ and expected degree $\langle k \rangle$, its excess degree distribution, i.e. the probability distribution of excess degree, is given by

$$Q(k) = \frac{(k+1)P(k+1)}{\langle k \rangle}$$

- (c) Express $g_1(x) = \sum_{k=0}^{\infty} Q(k)x^k$ in terms of $g_0(x) = \sum_{k=0}^{\infty} P(k)x^k$, where $g_0(x)$ and $g_1(x)$ are the generating function of $P(k)$ and $Q(k)$ respectively.
- (d) Calculate the excess degree distribution $Q_{\text{Poisson}}(k)$ which corresponds to a Poisson degree distribution, then calculate its generating function

$$G_1(x) = \sum_{k=0}^{\infty} Q_{\text{Poisson}}(k)x^k$$

Does it match the result of (c)?

- (e) Compare the results of (a) and (d). What can you observe?

Solution:

(a)

$$G_0(x) = e^{-c} \sum_{k=0}^{\infty} \frac{c^k x^k}{k!} = e^{-c+cx}$$

(b)

$$\begin{aligned} \langle k \rangle &= \left. \frac{dG_0(x)}{dx} \right|_{x=1} = c \\ \langle k(k-1) \rangle &= \left. \frac{d^2 G_0(x)}{dx^2} \right|_{x=1} = c^2 \end{aligned}$$

(c)

$$\begin{aligned} g_1(x) &= \frac{1}{\langle k \rangle} \sum_{k=0}^{\infty} (k+1)P(k)x^k = \frac{1}{\langle k \rangle} \sum_{k=0}^{\infty} kP(k)x^{k-1} \\ &= \frac{1}{\left. \frac{dG_0(x)}{dx} \right|_{x=1}} \frac{dG_0(x)}{dx} = \frac{g'_0(x)}{g'_0(1)} \end{aligned}$$

(d) Using the result of $\langle k \rangle$ in (b),

$$\begin{aligned} Q_{\text{Poisson}}(k) &= \frac{(k+1)P_{\text{Poisson}}(k+1)}{c} = \frac{k+1}{c} \frac{c^{k+1}e^{-c}}{(k+1)!} \\ &= \frac{1}{k!} c^k e^{-c} = P_{\text{Poisson}}(k) \end{aligned}$$

so $G_1(x) = G_0(x)$. This matches the result of (c), which suggests

$$G_1(x) = \frac{G'_0(x)}{G'_0(1)} = \frac{ce^{-c+cx}}{c} = e^{-c+cx} = G_0(x)$$

- (e) $P_{\text{Poisson}}(k) = Q_{\text{Poisson}}(k)$ and thus $G_0(x) = G_1(x)$. It means that the probability distributions of both the degree and excess degree are given by the same function in a random network. In fact, this is one of the explanations for why calculations are relatively easy on random networks.

(4) The average number of friends of your friends

Let $P(k)$ be the degree distribution of any network.

- (a) Show that

$$\frac{\langle k^2 \rangle}{\langle k \rangle} \geq \langle k \rangle$$

for all networks and that the equal sign holds only for regular networks.

- (b) The following paradox can be observed in many real social networks:

Your friends have more friends than you do.

Using the inequality in (a), show this paradox exists even in a randomly generated social network. (Hint: What is the probability for an edge to attach to a specific degree- k node? Then what is the probability for it to attach to any degree- k node?)

Solution:

- (a) Since the variance of degree is never negative,

$$\langle k^2 \rangle - \langle k \rangle^2 \geq 0 \quad \Rightarrow \quad \frac{\langle k^2 \rangle}{\langle k \rangle} \geq \langle k \rangle$$

The equal sign holds if and only if the variance of degree is 0, which happens when all nodes have the same degree or, in other words, when a network is regular.

- (b) Suppose the random network has n nodes and m edges. The probability for an edge to attach to a specific degree- k node reads $\frac{k}{2m-1} \approx \frac{k}{2m}$, then the probability for an edge to attach to any degree- k node is

$$q(k) = \frac{k}{2m} \times nP(k) = \frac{k}{\langle k \rangle} P(k)$$

Therefore, the average degree of a node's neighbor is

$$\sum_k kq(k) = \sum_k \frac{k^2}{\langle k \rangle} P(k) = \frac{\langle k^2 \rangle}{\langle k \rangle}$$

From (a), we have $\sum_k kq(k) > \langle k \rangle$, so the average degree of a node's neighbor is larger than the average degree of a node. In other words, your friends have more friends than you do.